Blockchain and IoT Technologies for Secure and Sustainable Urban Management: A Survey

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***Abstract*—The convergence of blockchain and Internet of Things (IoT) technologies presents transformative opportunities for secure and sustainable urban management. This survey synthesizes findings from three recent studies spanning blockchain based IoT security, blockchain for digital cultural heritage resources (DCHR), and IoT enabled smart parking systems. We analyze how decentralized architectures address persistent challenges in trust management, identity verification, data integrity, and resource optimization across these domains. The survey examines key enabling mechanisms including decentralized identifiers (DIDs), verifiable credentials (VCs), smart contracts, and lightweight consensus protocols. Comparative analysis reveals shared challenges of scalability, regulatory compliance, energy efficiency, and interoperability, alongside domain specific solutions. Results from the reviewed studies demonstrate that IoT based smart parking reduces search time by 42% and CO2 emissions by 27%, while blockchain based trust frameworks effectively mitigate Sybil attacks and device spoofing. We identify future research directions including AI blockchain fusion, green consensus algorithms, and cross domain standardization.**

***Index Terms*—blockchain, Internet of Things, trust management, decentralized identity, smart parking, cultural heritage, smart contracts, sustainability**

I. Introduction

The Internet of Things (IoT) ecosystem connects billions of heterogeneous devices ranging from smart home appliances and industrial sensors to healthcare monitors and autonomous vehicles. While this interconnected infrastructure enables unprecedented automation, efficiency, and real time decision making, it simultaneously introduces substantial challenges related to trust, security, identity management, and data integrity [1]. These challenges are amplified by the resource constrained nature of IoT devices, their physical exposure, and the dynamic topology of IoT networks.

Traditional centralized trust management approaches, including trusted third parties, centralized reputation systems, and cloud based access control, suffer from single points of failure, scalability limitations, and vulnerability to insider threats [1]. Blockchain technology, with its inherent properties of decentralization, immutability, transparency, and cryptographic security, provides a promising foundation for addressing these limitations across diverse application domains [2].

Beyond security, IoT technologies are increasingly deployed for urban sustainability objectives. Sensor equipped smart parking systems demonstrate measurable improvements in traffic congestion, fuel consumption, and carbon emissions through real time occupancy detection and predictive routing [3]. Similarly, blockchain has emerged as a transformative tool for digital cultural heritage preservation, where provenance tracking, authenticity certification, and intellectual property management are paramount concerns [2].

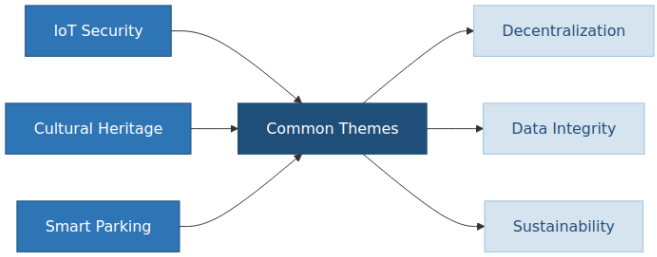

Fig. 1. Survey scope: three domains converging on common themes of decentralization, data integrity, transparency, and sustainability.

This survey synthesizes findings from three complementary studies: (1) a comprehensive review of blockchain for IoT security covering identity management, access control, and trust mechanisms [1]; (2) a review of blockchain technology for digital cultural heritage resources management [2]; and (3) a multiagent simulation study of IoT based smart parking for sustainable urban mobility [3]. By analyzing these works together, we identify cross domain patterns, shared challenges, and converging future directions at the intersection of blockchain and IoT technologies. Beyond synthesis, we present quantitative performance comparisons including authentication latency and transaction throughput from recent implementations, and conduct a cross domain security analysis mapping threat vectors to blockchain based mitigation mechanisms.

II. Literature Review

A. Blockchain for IoT Security

Khayer et al. [1] present a narrative review synthesizing 77 peer reviewed studies (2020–2025) on blockchain based trust management for IoT. Their PRISMA style screening process identified studies from IEEE Xplore, ACM Digital Library, SpringerLink, and MDPI, with quality assessment using a 0–8 rubric (median score: 5/8). The review spans

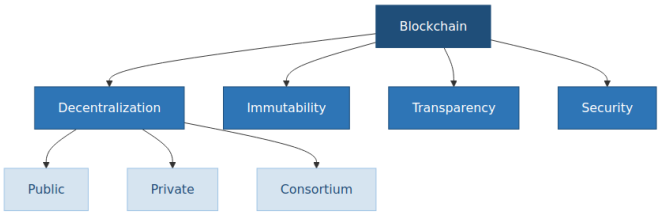

Fig. 2. Core blockchain properties and classification for IoT applications, showing how public, private, and consortium chains serve different security and governance requirements.

five major domains: industrial IoT (15 studies), smart cities (14), vehicular IoT (12), healthcare (13), and cross domain frameworks (23).

1) Trust Management Frameworks: Blockchain enables decentralized trust evaluation through device behavior analysis, interaction histories, and context aware mechanisms, replacing vulnerable centralized models. Notable approaches include integrating Physical Unclonable Functions (PUFs) with smart contracts for secure device identity [4], Sybil resistant trust models using blockchain based voting [5], and context aware trust modeling incorporating multi dimensional policy enforcement [6]. Bhatt and Sandhu [7] proposed an Attribute Based Access Control (ABAC) model for cloud enabled IoT, demonstrating fine grained control across smart home and smart parking scenarios.

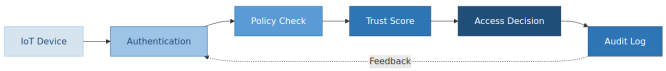

Fig. 3. Blockchain based trust management framework for IoT showing the flow from device authentication through trust evaluation to access decisions, with inputs from behavioral analysis, reputation scoring, and AI/ML models.

2) Decentralized Identity Management: Decentralized Identifiers (DIDs) provide globally unique identifiers controlled directly by the entity without requiring central registries. Dadkhah et al. [8] proposed a decentralized IoT onboarding framework for smart homes using consortium blockchain to ensure secure device integration. Ramírez-Gordillo et al. [9] introduced a decentralized identity management model using the IOTA Tangle to improve scalability in resource-constrained environments. Verifiable Credentials (VCs) enable selective disclosure of digitally signed credentials; Xu et al. [10] designed a privacy-preserving revocation mechanism using cryptographic accumulators to ensure tamper-evident credential management. Self Sovereign Identity (SSI) frameworks such as Hyperledger Indy and uPort grant devices full control over their digital identities [11].

3) Consensus Mechanisms for IoT: Traditional Proof of Work (PoW) consensus is impractical for resource constrained IoT devices. Ye et al. [12] proposed Proof of Trust (PoT), incorporating device reputation into consensus leader selection to reduce computational redundancy. Liu et al. [13] designed hierarchical blockchain architectures distributing consensus between edge and core layers. Hybrid approaches dynamically

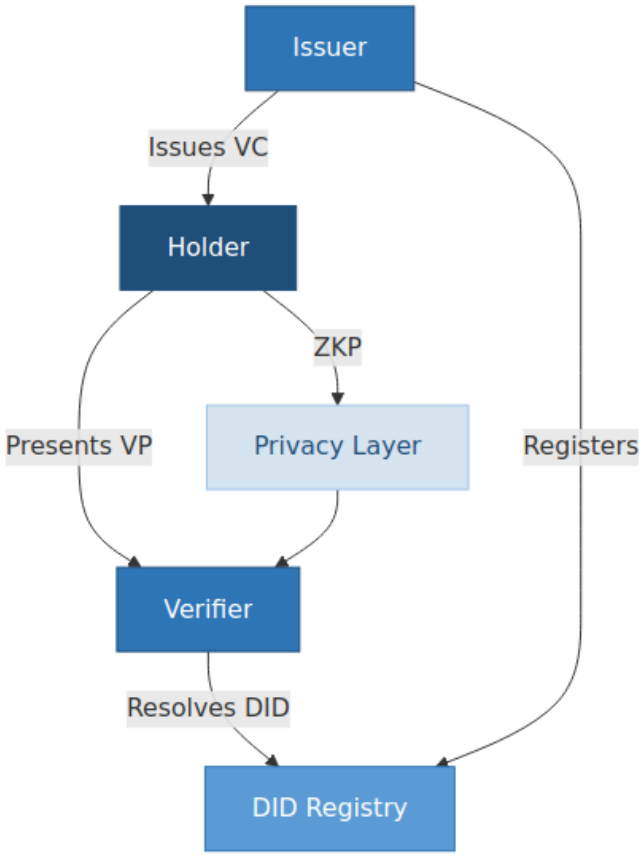

Fig. 4. Decentralized identity ecosystem showing the interactions among identity holders, credential issuers, and verifiers through DIDs, VCs, and privacy preserving mechanisms including zero knowledge proofs and selective disclosure.

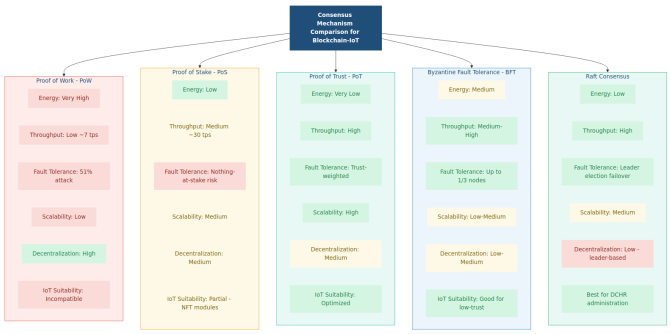

Fig. 5. Comparison of consensus mechanisms for blockchain IoT networks, evaluating each approach across energy efficiency, throughput, fault tolerance, scalability, and decentralization.

alternate between BFT and PoT based on node behavior and energy availability [14].

4) AI/ML Integration: AI and machine learning enhance blockchain IoT trust management through behavioral analysis, trust prediction, and anomaly detection. Federated intrusion detection systems integrate deep learning with blockchain for cross node intelligence sharing while preserving data privacy [15]. Reinforcement learning enables adaptive smart contract triggers for proactive resource management [16]. Hybrid deep learning frameworks combining bidirectional GRU and LSTM architectures have been proposed for dynamic trust assessment using device interaction histories, outperforming conventional static trust models [1]. AI driven anomaly detection systems combining artificial neural networks with adaptive regression on Hyperledger Fabric achieve real time detection with minimal computational overhead [1]. Fortino et al. [22] demonstrated a multi agent trust architecture where IoT devices operate as software agents with quantifiable reputation capital updated through blockchain certified interactions, successfully identifying nearly all adversaries when malicious agents remained below 25% of the network.

B. Blockchain for Digital Cultural Heritage

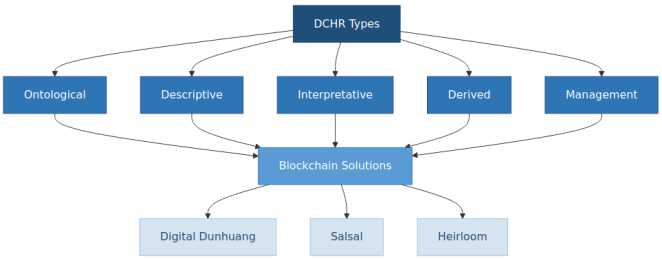

Fig. 6. Digital Cultural Heritage Resources taxonomy with five resource categories, blockchain solutions including NFT tokenization and smart contracts, and implementation cases from Digital Dunhuang to MApp.

Liu et al. [2] review blockchain applications for Digital Cultural Heritage Resources (DCHR), categorizing these resources into ontological, descriptive, interpretative, derived, and management types. Blockchain addresses key challenges in data authentication, privacy protection, copyright management, and equitable resource sharing.

The study evaluates consensus mechanisms specifically for DCHR, finding PoW incompatible due to high energy costs, PoS partially viable for NFT modules, and the Raft algorithm most suitable for centralized DCHR administration due to its fault tolerant leader election and strong consistency guarantees [2]. An enhanced Raft algorithm is proposed with improved security through tri phase mechanisms.

Two pivotal case studies demonstrate blockchain's practical impact: the Digital Dunhuang Open Material Library, which reconciles scalability with institutional collaboration through NFT driven economic models, and the Salsal platform, which embeds International Council of Museums ethical guidelines into Ethereum smart contracts for decentralized governance [2]. Additional platforms including Heirloom (IPFS based heritage preservation), MApp (minor artwork documentation across Ethereum and Polygon) [17], and BMDRM (NFT driven copyright management) illustrate the breadth of blockchain heritage applications.

The paper raises critical concerns about blockchain potentially establishing "technocratic authority" that transfers discursive power to developer communities, and about NFT commodification conflicting with communal ownership norms of indigenous heritage [2].

C. IoT for Smart Urban Mobility

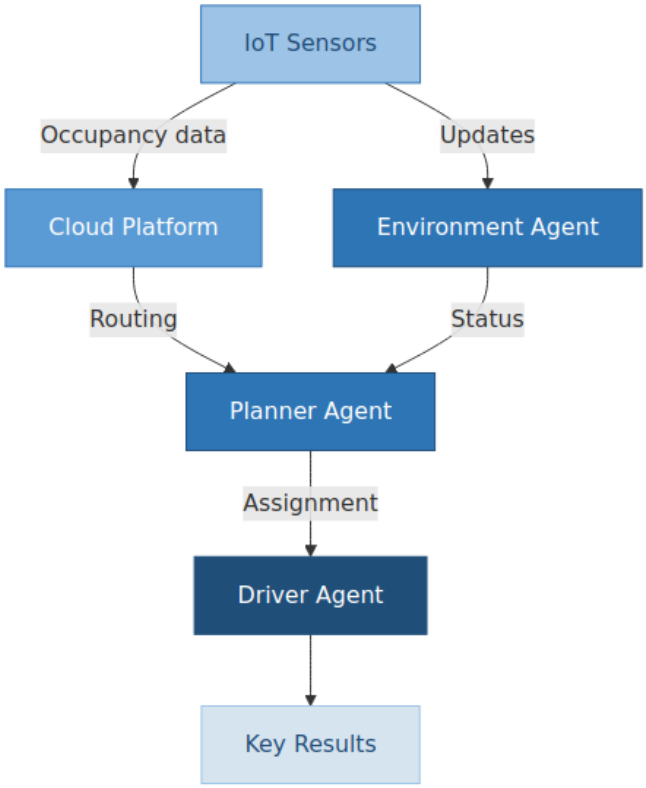

Fig. 7. IoT smart parking system architecture showing three agent types (driver, planner, environment), sensor based occupancy detection, cloud analytics, and key performance results.

Mutambik [3] constructs a multiagent simulation framework using the Janus platform to evaluate IoT based smart parking. The system employs ultrasonic, magnetic, and infrared sensors for real time occupancy detection, with a predictive guidance module computing optimal routes using real time and historical data.

The simulation models three agent types using Finite State Machine (FSM) transitions: driver agents (idle/navigate/park states), planner agents (coordination and assignment), and environment agents (infrastructure and status updates). Testing across 200–500 parking units with 30 repetitions per scenario over simulated one year periods, the framework evaluates performance across four dimensions [3].

Results demonstrate substantial improvements: average parking search time decreased by 42%, congestion within

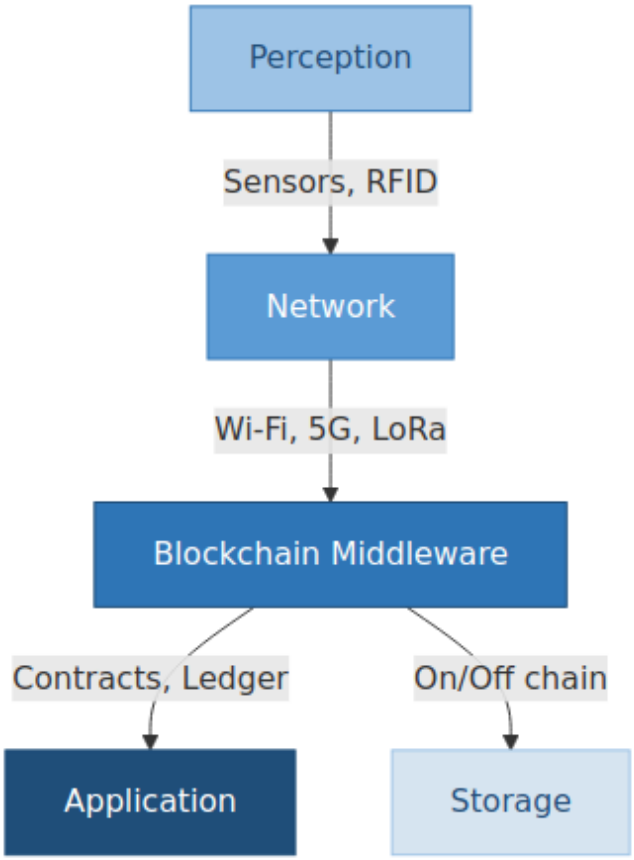

Fig. 8. IoT architecture layers with blockchain integration, spanning perception (sensors), network (communication protocols), middleware (blockchain), application (smart city services), and storage (on chain and off chain).

parking zones dropped by 35%, and CO2 emissions from circling vehicles reduced by 27% [3]. The smart system reduced daily travel distance from approximately 384 km (periodic weekly strategy) to 71 km (smart threshold 0.9), with corresponding CO2 reductions from 309 kg/day to below 75 kg/day. Regarding time efficiency, the periodic strategy with weekly reviews required up to 15.7 hours of cumulative daily search time, while the smart threshold 0.9 system reduced this to approximately 3.5 hours, a 77% reduction during peak periods. User satisfaction measurements indicated that overutilized parking zones per day decreased from 3.2 under the periodic strategy to 2.16 under the smart system. Sensitivity analysis across 200 to 500 parking units confirmed that the smart model maintained consistently shorter travel distances across all configurations, while the periodic model's distances increased continuously with zone numbers [3]. Computational performance of 1 to 2 seconds per simulated day for 200 units, with execution time scaling linearly, indicates favorable scalability for real time deployment.

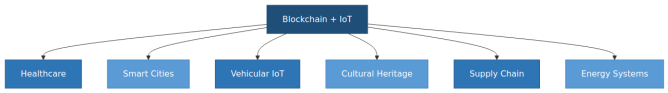

Fig. 9. Cross domain applications of blockchain and IoT technologies spanning healthcare, smart cities, vehicular IoT, cultural heritage, supply chain, and energy systems, supported by shared enabling technologies.

III. Analysis and Comparison

A. Cross Domain Analysis

Table I presents a comparative analysis of the three surveyed studies across key dimensions.

B. Shared Challenges and Solutions

Table II maps shared challenges across the three domains to their proposed solutions.

C. Research Gaps

Analysis reveals several cross domain research gaps. First, all three studies acknowledge the absence of large scale real world deployments; most blockchain IoT approaches rely on prototypes or simulations. Second, quantitative benchmarking across solutions remains limited; the IoT security survey notes that no study directly compares consensus mechanisms under identical conditions. Third, the economic viability of blockchain deployment, including total cost of ownership and return on investment, is insufficiently explored. Fourth, the smart parking study identifies the need for blockchain integration to secure sensor data transactions, representing a natural convergence point between the IoT security and urban mobility domains.

D. Performance Results: Throughput and Latency

The surveyed studies provide quantitative performance evidence illuminating the practical viability of blockchain IoT integration. Table III summarizes key metrics reported across the reviewed literature.

In the blockchain IoT security domain, authentication latency and transaction throughput are critical metrics. Mazzocca et al. [20] developed EVOKE, an IoT oriented verifiable credential revocation scheme using elliptic curve based cryptographic accumulators, reporting approximately 1.5 KB overhead per node and witness generation of roughly 80 ms per million VCs, demonstrating scalable credential lifecycle management with minimal resource impact.

For resource constrained devices, Fotiou et al. [21] demonstrated a capabilities based access control framework encoding fine grained rights within W3C VCs, with prototype evaluations on Raspberry Pi and ESP32 showing VC generation under 0.1 ms and on device verification in 10 to 160 ms. Healthcare IoT pilots validated blockchain based key management reducing latency by 20% compared to traditional PKI [1]. Liu et al. [13] showed that hierarchical blockchain architectures distributing consensus between edge and core layers achieve significant improvements in both latency and throughput through local lightweight validation at the edge.

TABLE I
COMPARATIVE ANALYSIS OF SURVEYED STUDIES

Dimension	Blockchain for IoT Security [1]	Blockchain for DCHR [2]	IoT Smart Parking [3]
Primary Focus	Trust, identity, access control	Heritage preservation, provenance	Urban mobility, sustainability
Blockchain Role	Decentralized trust and identity	Immutable provenance and rights	Future integration for security
IoT Role	Target environment (devices)	Environmental sensing for artifacts	Core sensing infrastructure
Consensus	PoT, adaptive PBFT, hybrid	Enhanced Raft, BFT	Not applicable
Smart Contracts	Access control, trust updates	Copyright, ethical governance	Not implemented
Evaluation	Qualitative survey (77 studies)	Qualitative review (86 refs)	Quantitative simulation
Validation	Literature synthesis	Case studies	Agent based simulation
Key Metrics	Latency, throughput, energy	Qualitative assessments	Distance, CO2, time, satisfaction

TABLE II
SHARED CHALLENGES ACROSS DOMAINS

Challenge	Solutions Across Domains
Scalability	Hierarchical blockchains [13], edge aggregation, IPFS offloading [17], linear simulation scaling [3]
Energy	PoT consensus [12], enhanced Raft [2], sensor duty cycling [3]
Regulatory	Off chain storage with hash anchoring [1], GDPR compliant consent management [18], cross border sandboxes [2]
Interop.	Standardized DIDs/VCs [11], ERC-1155 cross chain [17], open APIs
Privacy	Zero knowledge proofs [19], selective disclosure, differential privacy [3]

In the smart parking domain, the predictive routing module reduced cumulative daily search time from 15.7 hours to 3.5 hours, a 77% reduction translating to improved response latency in driver system interactions [3]. For digital cultural heritage, IPFS integration for off chain storage significantly reduces on chain transaction size, with MApp managing over 15,000 minor artwork records across Ethereum and Polygon chains [17].

TABLE III
PERFORMANCE METRICS ACROSS BLOCKCHAIN IOT SOLUTIONS

Solution	Metric	Result
EVOKE [20]	VC overhead	1.5 KB per node
EVOKE [20]	Witness gen.	80 ms per million VCs
CapBAC [21]	VC generation	$<$0.1 ms
CapBAC [21]	Verification	10–160 ms (RPi/ESP32)
Hierarchical [13]	Throughput	Improved via edge/core
Smart parking [3]	Search time	15.7 h $\rightarrow$ 3.5 h
MApp [17]	Records	15,000+ artworks tracked

These results demonstrate that lightweight consensus, hierarchical architectures, and off chain storage enable blockchain IoT systems to meet practical latency and throughput requirements on constrained devices. However, standardized benchmarking frameworks are needed, as most studies evaluate under different testbed conditions [1].

E. Security Analysis

Security remains a fundamental concern across all three surveyed domains. This section analyzes threat models, mitigation mechanisms, and residual vulnerabilities.

1) Threat Landscape: The blockchain IoT ecosystem faces diverse threats varying by domain. Khayer et al. [1] identify primary attack vectors: Sybil attacks where adversaries create fake identities to manipulate trust assessments; device spoofing through unauthorized impersonation; insider threats from compromised administrators; data tampering targeting sensor readings and transactions; and man in the middle attacks intercepting communications. For cultural heritage, threats include provenance forgery, unauthorized modification of artifact records, and double spending in NFT transactions [2]. Smart parking systems face risks of sensor data manipulation and unauthorized access to occupancy information [3].

2) Mitigation Mechanisms: Blockchain provides multiple security layers across domains. Decentralization eliminates single points of failure, distributing risk across participants [1]. Immutability ensures recorded data cannot be retroactively altered, providing tamper evident audit trails for heritage provenance [2] and IoT device logging.

Physical Unclonable Functions (PUFs) integrated with smart contracts provide hardware rooted authentication, countering Sybil attacks and device cloning. Javaid et al. [4] demonstrated this for vehicular IoT, binding blockchain identities to unique hardware characteristics. Sybil resistant models using blockchain based voting strengthen resilience through distributed reputation validation [5].

Zero knowledge proofs enable privacy preserving authentication without revealing sensitive information. The zk IoT framework [19] maintains encrypted access logs on chain while preserving anonymity. Smart contracts automate security policy enforcement, with ABAC models enabling dynamic permissions based on device attributes and behavioral profiles [7]. Federated intrusion detection combining deep learning with blockchain enables collaborative threat detection while preserving data privacy [15].

Table IV maps primary threats to mitigation mechanisms across domains.

TABLE IV
SECURITY THREAT MITIGATION ACROSS DOMAINS

Threat	Mitigation	Domain
Sybil attacks	PUF + blockchain voting	IoT security [4], [5]
Device spoofing	Smart contract verification	IoT security [4]
Data tampering	Immutable ledger	All domains [1], [2]
Privacy violation	ZKPs, selective disclosure	IoT, DCHR [2], [19]
Insider threats	Decentralized ABAC	IoT security [7]
Provenance forgery	NFT tokenization	DCHR [2], [17]
Sensor manipulation	Not yet addressed	Smart parking [3]

3) Residual Vulnerabilities: Despite these advances, significant gaps persist. The smart parking study explicitly acknowledges absent security mechanisms, identifying blockchain integration for secure sensor transactions as a critical future need [3]. The DCHR domain faces challenges where immutability may freeze contested heritage narratives, and the enhanced Raft algorithm remains vulnerable to sustained attacks on predefined nodes [2]. Across all domains, tensions between blockchain transparency and GDPR privacy requirements remain partially unresolved, with current off chain storage and consent management patterns adding architectural complexity [1], [18].

IV. CONCLUSION

This survey synthesizes three complementary studies to examine how blockchain and IoT technologies address challenges in security, cultural heritage preservation, and urban sustainability. Our analysis identifies several key findings.

Performance evaluation reveals that blockchain IoT solutions achieve practical throughput and latency levels, with verifiable credential operations executing in under 160 ms on constrained hardware [21]. Security analysis demonstrates that PUFs, zero knowledge proofs, ABAC smart contracts, and federated intrusion detection effectively address major threats including Sybil attacks, device spoofing, and insider threats, though cross domain security integration and regulatory compliance remain open challenges [1].

Blockchain provides robust mechanisms for decentralized trust management through DIDs, VCs, and smart contracts, effectively mitigating threats including Sybil attacks, device spoofing, and insider threats in IoT environments. For cultural heritage, blockchain enables immutable provenance tracking, NFT based tokenization, and ethical governance through smart contract encoded institutional guidelines. IoT sensor networks, when combined with predictive analytics, demonstrate measurable sustainability improvements: 42% reduction in parking search time, 35% congestion decrease, and 27% lower CO2 emissions.

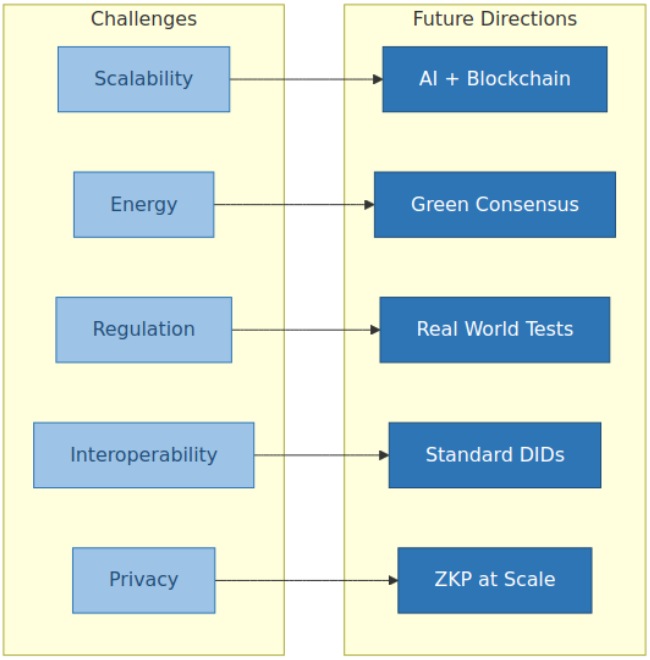

Fig. 10. Mapping of current challenges to emerging solutions and future research directions, showing progression from scalability and energy issues toward AI blockchain fusion and green consensus protocols.

Shared challenges across domains include scalability limitations of public blockchains, regulatory tensions between immutability and privacy regulations such as GDPR, energy consumption of traditional consensus mechanisms, and interoperability across heterogeneous systems. Promising solutions include hierarchical blockchain architectures, lightweight consensus protocols such as Proof of Trust, off chain storage patterns, and standardized identity frameworks.

Future research should pursue several converging directions: (1) AI blockchain fusion through federated learning with on chain triggers for privacy preserving intelligence; (2) green consensus algorithms designed for constrained IoT environments; (3) large scale real world validation across diverse urban contexts; (4) cross domain standardization of DID schemas and onboarding protocols; and (5) regulatory frameworks that reconcile decentralized architectures with evolving privacy legislation. The convergence of blockchain and IoT technologies holds substantial promise for building secure, transparent, and sustainable urban management systems, provided that scalability, compliance, and deployment cost barriers are systematically addressed.

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